

# Cartwheel Homework 6

Repeatable agent evaluations in Harbor and GitHub Actions

|                                           |                                        |                                         |                                                   |
|-------------------------------------------|----------------------------------------|-----------------------------------------|---------------------------------------------------|
| <div>11</div> <div>classified cases</div> | <div>55</div> <div>baseline runs</div> | <div>110</div> <div>CI agent runs</div> | <div>15</div> <div>capability analysis runs</div> |
|-------------------------------------------|----------------------------------------|-----------------------------------------|---------------------------------------------------|

**Outcome**

We converted two observed Homework 4 failure modes into 11 trace-level cases, established five-run baselines, implemented pass@k and pass^k, and ran two full CI evaluations from one pull request. The intentional email-change regression failed 0/5 and recovered to 5/5 after its prompt instruction was removed.

Model: **gpt-5.6-luna** - Service tier: **Fast** - Judge calls: **0** - Final local commit: **0a9d6a1** - Report date: **September 26, 2026**

This screenshot report replaces the optional walkthrough video for the student's course record. The committed cases, Harbor result summaries, CI URLs, and capability analysis remain the authoritative evidence.

# 1. Cases and five-run baselines

## Failure mode 1: missing escalation

Cases e-001 through e-005 cover account changes, charge disputes, and payment credential changes. Case e-011 probes an invalid negative product price that should trigger human escalation.

## Failure mode 2: missing policy grounding

Cases e-006 through e-010 require the correct help-center lookup and exact policy identifier. All decisions use deterministic tool-call and reply-content checks.

| Cases          | Mode                        | Baseline | Classification |
|----------------|-----------------------------|----------|----------------|
| e-001 to e-005 | Missing required escalation | 5/5 each | Regression     |
| e-006 to e-010 | Missing policy grounding    | 5/5 each | Regression     |
| e-011          | Missing required escalation | 0/5      | Capability     |

## Why e-011 is a capability

All five baseline replies noticed that the seeded -\$5.00 price was invalid, but none called `escalate_to_human`. The desired behavior was therefore not reliable enough to protect as a regression.

## Scope and cost controls

The final set contains 11 cases, 55 baseline agent runs, and no LLM judge calls. Exact facts are checked in code, and the rejected Homework 5 judge is not used for autonomous enforcement.

# 2. Reliability metrics and CI contract

| Component      | Behavior                                                                                                    |
|----------------|-------------------------------------------------------------------------------------------------------------|
| pass@k         | Estimates the chance of at least one success in k attempts from n observed runs with c successes.           |
| pass^k         | Estimates the chance that all k attempts succeed. This is the reliability metric used for regression cases. |
| case_passes    | Blocks CI when any completed regression trial fails. Capability cases report metrics without blocking CI.   |
| Harbor adapter | Reads Harbor 0.23 per-trial result files while retaining compatibility with the earlier aggregate layout.   |

## GitHub Actions

Every pull request installs Harbor 0.23, generates the classified tasks, runs five Docker-isolated attempts per case, summarizes even after failures, and uploads the full Harbor job artifact. CARTWHEEL\_MODEL is a repository variable; provider credentials remain encrypted secrets.

## Validation

The completed branch passed 86 offline checks locally. The pass metric and Harbor adapter subset passed 40 checks. A timing-sensitive assertion on an unrelated SDK warning was removed while preserving the direct no-export assertion; the final GitHub offline workflow passed.

Final free validation: [github.com/SCRKT/cartwheel-homeworks/actions/runs/36236561526](https://github.com/SCRKT/cartwheel-homeworks/actions/runs/36236561526). The Harbor evaluation job was skipped because the demonstration pull request had been closed after its two required paid runs.

# 3. One pull request, two evaluation commits

The public fork records the complete HW6 evaluation pipeline. Commit 7942fc6 adds the intentional e-001 regression; commit 85084d2 removes it on the same pull request. The PR was closed after both required paid runs to prevent later artifact-only pushes from starting a third batch.

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## Complete HW6 Harbor CI evaluation #1

🔒 Closed

SCRKT wants to merge 7 commits into `main` from `hw6-ci-clean`

💬 Conversation 0

📄 Commits 7

🔍 Checks 4

📄 Files changed 8

+298 -26

SCRKT commented 7 hours ago

Owner ...

Adds the Homework 6 evaluation pipeline and turns reviewed Homework 4 failures into repeatable Harbor cases.

- Adds 11 cases across missing escalation and missing policy grounding, classified from five baseline attempts each.
- Implements `pass@k`, `pass^k`, and regression-versus-capability CI decisions.
- Supports Harbor 0.23's per-trial result layout and adds regression coverage.
- Runs five attempts per case in GitHub Actions with `gpt-5.6-luna` on the Fast service tier.
- Includes the temporary `e-001` email-change regression required for the first Part D run. A follow-up commit on this PR will revert it for the second run.

Validation: 86 passed, 14 skipped, 13 xfailed, 3 xpassed .

👤 Samuel Crockett added 7 commits 8 hours ago

📄 Draft HW6 baseline evaluation cases

61137f1

Development

📄 Support Harbor 0.23 per-trial results

b8bec29

Successfully merging this pull request may close these issues.

📄 Classify initial HW6 cases and add capability probe

fe9dc7f

None yet

📄 Complete HW6 baselines and pass metrics

f14a286

1 participant

📄 Configure HW6 Harbor evaluation workflow

d38a22c

📄 Introduce HW6 email escalation regression

✖ 7942fc6

📄 Revert intentional HW6 email regression

✖ 85084d2

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# 4. Intentional regression run

The first pull-request run completed all 55 Harbor trials and uploaded one job artifact. The workflow failed at the summary step because e-001 scored 0/5 and was correctly classified as a blocking regression. Offline checks passed. Offline checks passed.

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Complete HW6 Harbor CI evaluation #2

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Summary

All jobs

offline checks

Harbor evaluations

Run details

Usage

Workflow file

Triggered via pull request 7 hours ago

Status

Total duration

Artifacts

SCRKT opened #1 hw6-ci-clean

Failure

16m 32s

1

evals.yml

on: pull\_request

offline checks 21s

Harbor evaluations 16m 28s

Annotations

1 error, 2 warnings, and 2 notices

Harbor evaluations

Process completed with exit code 1.

offline checks

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, astral-sh/setup-uv@v5. For m...

Show more

Harbor evaluations

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/upload-artifact@v4, ...

Show more

# 5. Reverted regression run

The second run used the corrected prompt. e-001 recovered to 5/5. The workflow remained red because e-003 and e-006 each produced one behavioral failure; these completed outcomes were retained as new evidence. The separate offline failure was a timing-sensitive logging assertion fixed and validated afterward.

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Complete HW6 Harbor CI evaluation #4

Sign in to view logs

Summary

All jobs

offline checks Harbor evaluations

Run details Usage Workflow file

Triggered via pull request 6 hours ago

SCRKT synchronize #1 hw6-ci-clean

Status Failure

Total duration 13m 12s

Artifacts 1

evals.yml

on: pull\_request

offline checks 24s

Harbor evaluations 13m 7s

Annotations

2 errors, 2 warnings, and 2 notices

offline checks

Process completed with exit code 1.

Harbor evaluations

Process completed with exit code 1.

offline checks

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, astral-sh/setup-uv@v5. For m...

Show more

Harbor evaluations

# 6. CI results and interpretation

| Case             | First run   | After revert | Interpretation                    |
|------------------|-------------|--------------|-----------------------------------|
| e-001            | 0/5 - block | 5/5 - pass   | Intentional regression recovered  |
| e-002            | 5/5         | 5/5          | Reliable in both runs             |
| e-003            | 5/5         | 4/5 - block  | New reliability failure           |
| e-004            | 5/5         | 5/5          | Reliable in both runs             |
| e-005            | 5/5         | 5/5          | Reliable in both runs             |
| e-006            | 3/5 - block | 4/5 - block  | Baseline reliability did not hold |
| e-007 to e-010   | 5/5 each    | 5/5 each     | Reliable in both runs             |
| e-011 capability | 2/5         | 1/5          | Reported, never blocks            |

## Run 1

Commit 7942fc6 - 55 agent runs - 0 judge calls - 16m32s - artifact uploaded. URL: [github.com/SCRKT/cartwheel-homeworks/actions/runs/36215130310](https://github.com/SCRKT/cartwheel-homeworks/actions/runs/36215130310)

## Run 2

Commit 85084d2 - 55 agent runs - 0 judge calls - 13m12s - artifact uploaded. URL: [github.com/SCRKT/cartwheel-homeworks/actions/runs/36216043198](https://github.com/SCRKT/cartwheel-homeworks/actions/runs/36216043198)

# 7. Capability case after 5, 10, and 15 runs

| Observed runs | Successes | pass@1 | pass@3 | pass@5 | pass@10 | pass@15 |
|---------------|-----------|--------|--------|--------|---------|---------|
| 5             | 2         | 0.400  | 0.900  | 1.000  | -       | -       |
| 10            | 4         | 0.400  | 0.833  | 0.976  | -       | -       |
| 15            | 4         | 0.267  | 0.637  | 0.846  | 0.996   | 1.000   |

## Stability conclusion

The final five trials all failed, moving pass@1 from 0.400 to 0.267, pass@3 from 0.833 to 0.637, and pass@5 from 0.976 to 0.846. These changes are material, so 15 observed runs did not produce a stable estimate. A product should not rely on repeated attempts to compensate for the missing escalation behavior.

| Artifact            | Location or result                                                                     |
|---------------------|----------------------------------------------------------------------------------------|
| Cases               | eval_cases/cases.jsonl - 11 classified cases across two failure modes                  |
| Baselines           | eval_results/hw6-baseline.json - 55 completed trials, zero exceptions                  |
| CI evidence         | ci-runs.json - both run URLs, commits, targeted outcomes, and explanation              |
| Capability analysis | eval_results/e-011-15.json - ordered trials and 5/10/15 estimates                      |
| Workflow            | .github/workflows/evals.yml - Harbor 0.23, Docker, five attempts, summary and artifact |
| Fork branch         | SCRKT/cartwheel-homeworks:hw6-ci-clean - final commit ed01b9b                          |